



Incident report analysis

Summary	This afternoon, several critical components of the company's internal network suddenly stopped functioning. Shortly afterwards the IT department investigated the incident and found that a high volume of ICMP packets were sent to the network, overwhelming the servers' capacity. IT also discovered that the firewall was improperly configured and failed to stop the flood attack from occurring. The team responded by blocking incoming ICMP packets, and then taking less critical sections of the network offline. Critical services were restored to the network, and baseline checks were performed to ensure that data was not manipulated. As a result of the downtime, customers were unable to access all multimedia services hosted by the company for a short time.
Identify	The incident management and response team conducted a network audit following the incident to create a strategy for reducing the network's attack surface in the future. More specifically, the team discussed how to prevent and respond to denial of service and other flood attacks. The primary vulnerability was the misconfiguration of the firewall which failed to limit incoming traffic.
Protect	Several patches were made to the network, including a new firewall rule to limit the rate of incoming ICMP packets, new network monitoring software, and an IDS/IPS system to guard against ICMP flood attacks in the future. Source IP verification was also implemented to reduce the system's vulnerability to DoS attacks.
Detect	The intrusion detection and prevention software that was installed by our team aims to greatly reduce the risk of unwanted traffic reaching the target network and overwhelming it. Suspicious traffic flagged by the system is automatically

	filtered out to ensure that a reasonable amount of ICMP packets reach the network. The addition of source IP verification also ensures that requests to the network come from legitimate devices.
Respond	As soon as the incident response team was made aware of the incident, we disabled less critical sections of the network and focussed on restoring essential functionality. Then the entire system was set back online and further investigation was conducted to determine which services were targeted and which security measures would best prevent the incident from happening again. Patches mentioned above were implemented and the network was restored to a working and secure condition.
Recover	Further analysis was conducted to see if data was corrupted or manipulated, however it was concluded that the attack did not alter any code or software/hardware configurations. The only real consequence of the security incident was the time the server was offline and multimedia services could not adequately be provided to customers.

Reflections/Notes: